27/02/2025

Exercise Session - no. 2

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AnW - 2025

Every graph has an even number of odd degree vertices.

- ○ True
- ○ False

Every graph has an even number of odd degree vertices.

- (selected) True
- False

Every articulation point in a graph is incident to a bridge.

- ○ True
- ○ False

Every articulation point in a graph is incident to a bridge.

○ True

◉ False

There is a 5-connected graph with all vertices of degree exactly 3.

- ○ True
- ○ False

There is a 5-connected graph with all vertices of degree exactly 3.

- [ ] True
- [x] False

A Hamiltonian cycle in a graph visits every vertex exactly once.

- ○ True
- ○ False

A Hamiltonian cycle in a graph visits every vertex exactly once.

- (selected) True
- False

For two edges a, b of a graph, the relation $a \sim b$ when a and b are on a common cycle is an equivalence relation.

- ○ True
- ○ False

For two edges a, b of a graph, the relation $a \sim b$ when a and b are on a common cycle is an equivalence relation.

- [x] True
- [ ] False

Every connected graph with all vertices of even degree has an Eulerian cycle.

- ○ True
- ○ False

Every connected graph with all vertices of even degree has an Eulerian cycle.

- [x] True
- [ ] False

Every 2-connected graph has a Hamiltonian cycle.

- ○ True
- ○ False

Every 2-connected graph has a Hamiltonian cycle.

- ( ) True
- (•) False

One reason -> P vs. NP

Complexity to find cut vertices?

If $G = (V, E)$ is a 3-connected graph and $v \in V$, then $G[V \setminus \{v\}]$ is 2-connected.

- ○ True
- ○ False

If $G = (V, E)$ is a 3-connected graph and $v \in V$, then $G[V \setminus \{v\}]$ is 2-connected.

- [x] True
- [ ] False

For every $t \geq 3$ a complete graph K_t on t vertices is 2-connected.

- ○ True
- ○ False

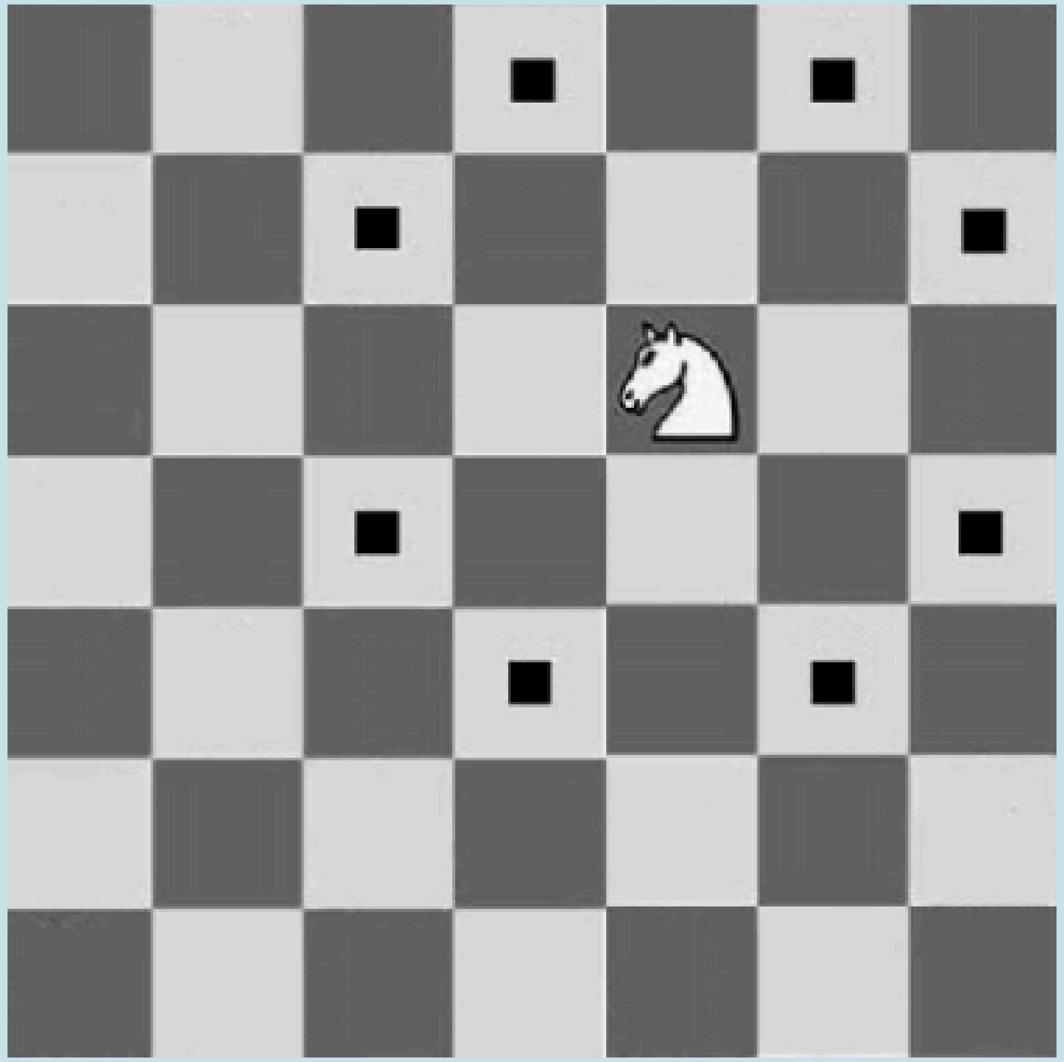
For every $t \geq 3$ a complete graph K_t on t vertices is 2-connected.

- [x] True
- [ ] False

K_t => Hamilton cycle => 2-connected

There is a Hamiltonian cycle for a chess knight jumping on 7×7 chessboard.

As a reminder, legal chess knight moves are shown below.



- ○ True
- ○ False

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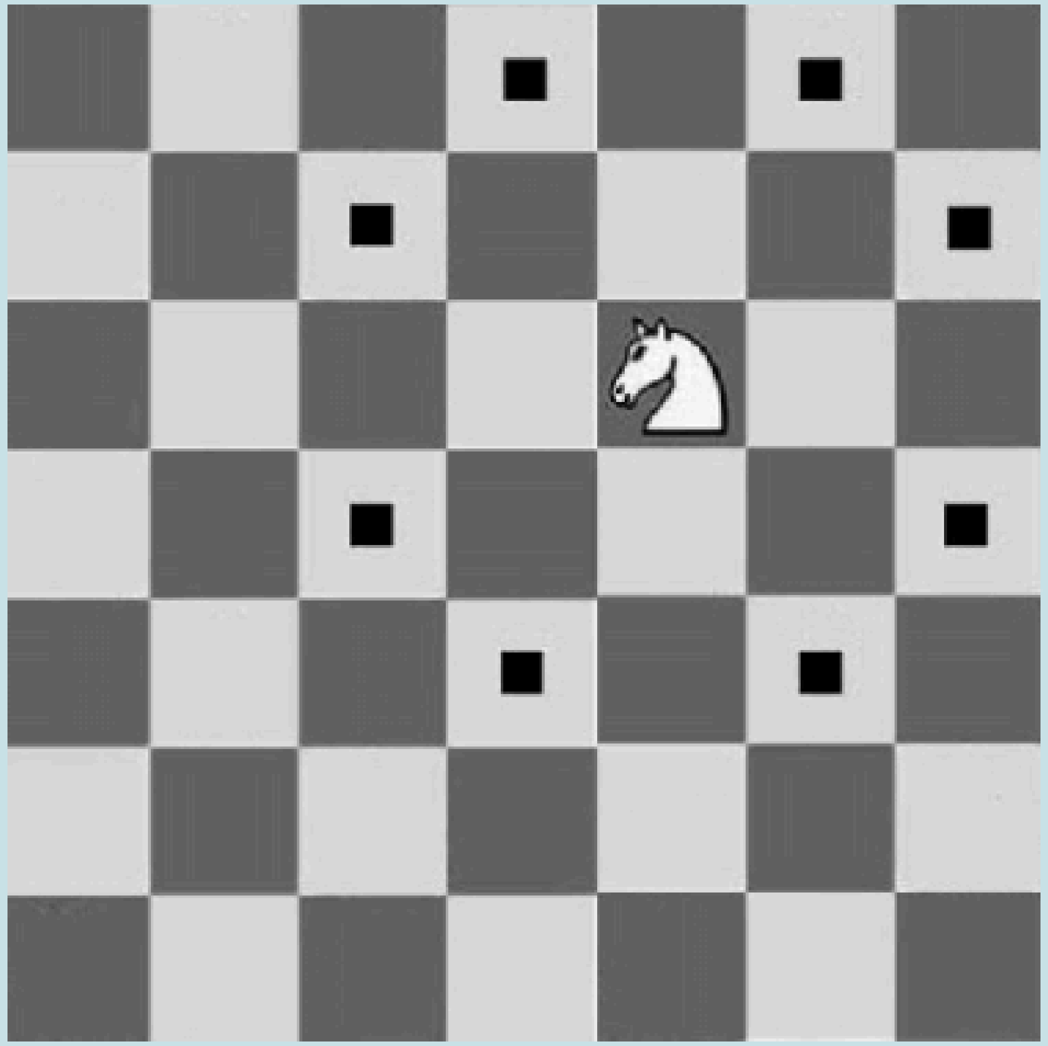


- [ ] True
- [x] False

Lecture Recap

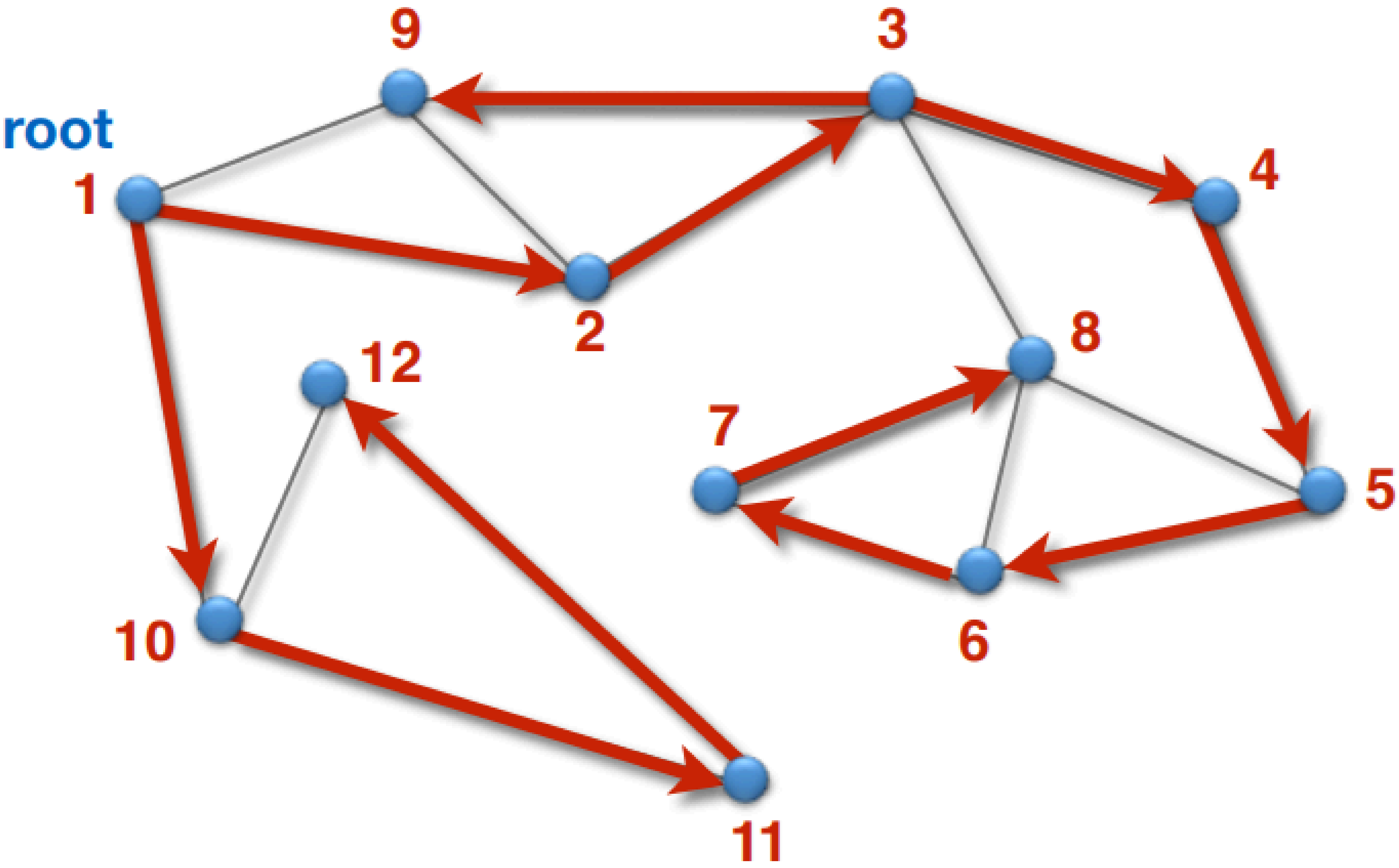
How to find all cut vertices in an efficient way?

How to find all cut vertices in an efficient way?

Let G = (V,E) an undirected graph. Assume G is connected

1. *What is DFS? What about the DFS-tree?*
2. *We can now classify all edges in 2 categories*
   a. *Tree edges (Baumkanten)*
   b. *Unused edges (Restkanten)*
3. *Types of unused edges in directed vs undirected graph?*

!! If there is no direction in undirected graphs, how can I say if one edge is back-edge or forward edge?


root
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How to find all cut vertices in an efficient way?

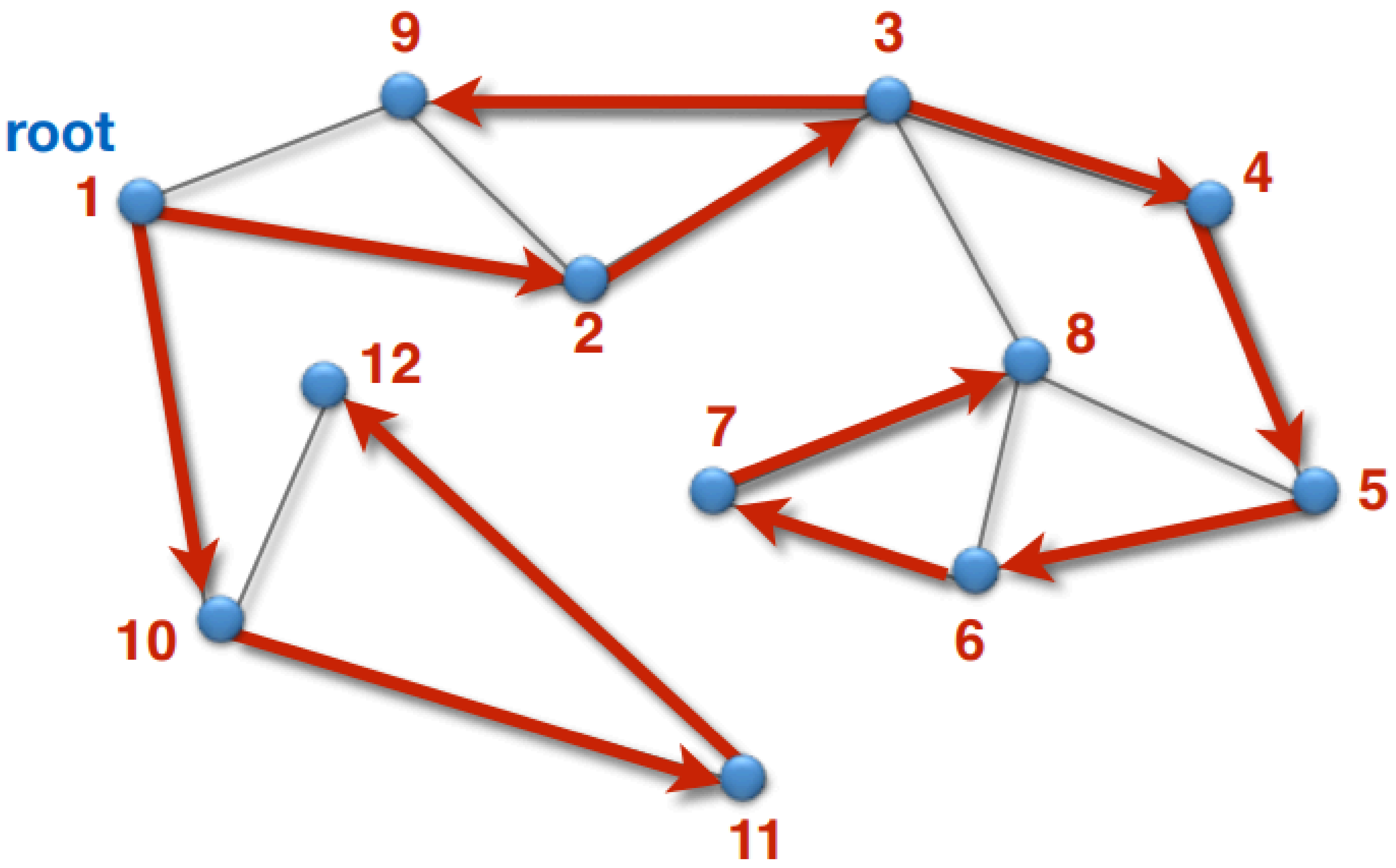
Let G = (V,E) an undirected graph. Assume G is connected

Intuition...

- *What do we know about our graph if there are cut vertices? (+ the fact that it is (1)-connected?)*
- *Is our graph 2-connected?*
- *When is a graph 2-connected? (Menger's Theorem)*
- *What is this saying about my cut vertex w? .... Now I know that for some (u,v), the only way to connect the nodes is using w... i.e, there is no cycle containing u and v*

So... if some vertex have this node on their path to another one... and that is the only path available, after removing the cut-vertex, the 2 nodes will be in dif. connected componets.

Back to our DFS-tree ...

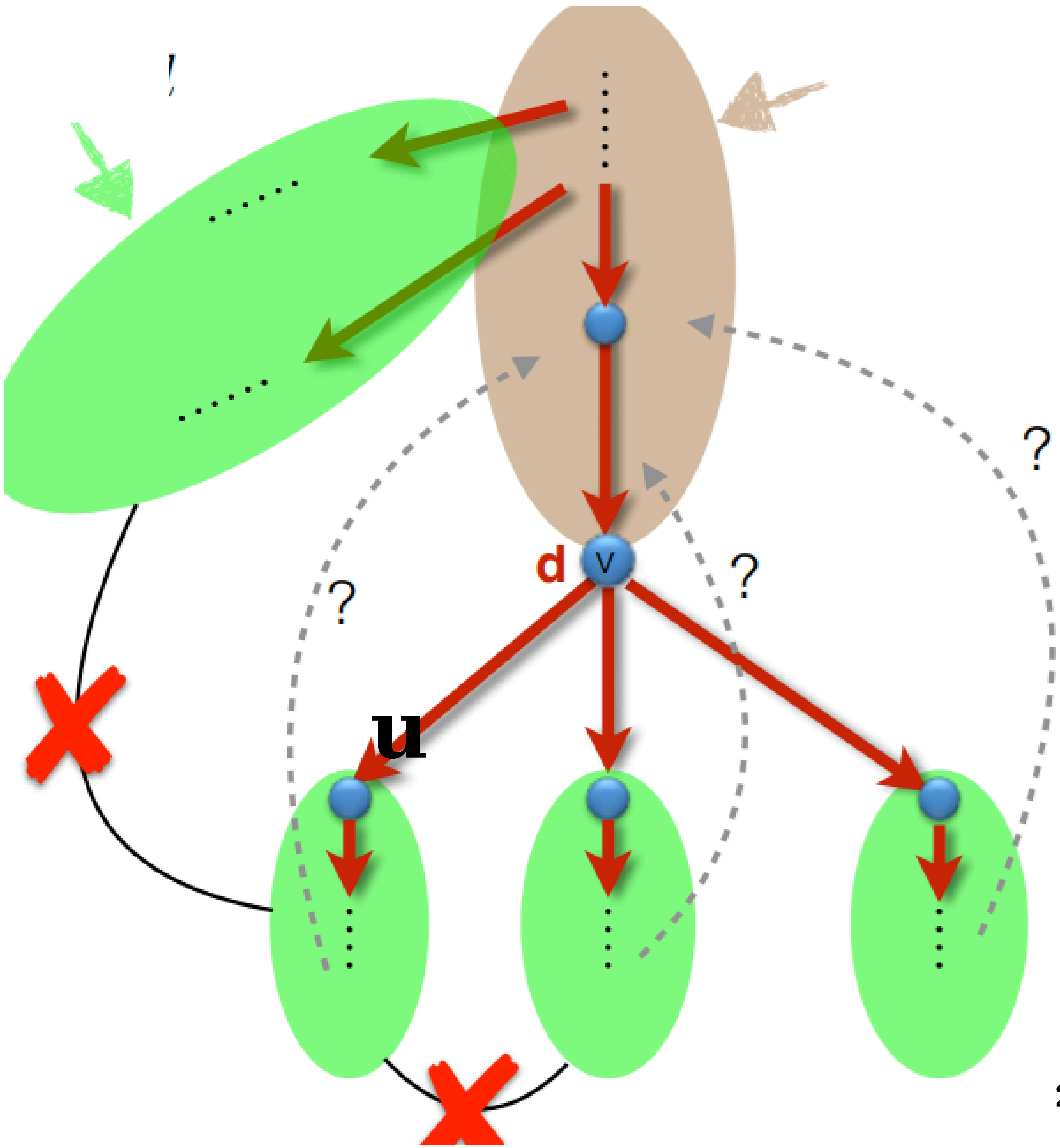


- for a node u, that is a child of v, it means that u is using v on one of its path

- Is there another path from an upper node\ from other subgraph to u that is not using v?

- iff there is a cycle containing u that starts "before v" (i.e. {v, u} is included) => see next slide

Back to our DFS-tree …



We are done with the intuition part… what about a formal solution for all of this…

lor all v in V, we define low[v] as...

low[v] := the smallest dfs number that can be reach starting from the node v in the dfs-tree, such that we are allowed to use as many tree edges we want, but only one unused edge ...

.... ofc, we will use a back-edge. Using a forward edge will only increase the low[v] (since all the nodes under v have a larger dfs number

Computing the low numbers for all v in V…. but how?

DP approach … 🤓

$$low[v] = min(dfs[v], \min_{(v,w)\in E} \begin{cases} low[w], & (v,w) tree-edge \\ dfs[w], & (v,w) rest-edge \end{cases})$$

only one problem left… what is the order in which one should compute the entries of the DP table?

dfs is already computed… so I want all low[w] to be compute as well for all (v, w)… if (v, w) is a tree-edge, what do I know about v and w ?

Computing the low numbers for all v in V.... but how?

What about the time-complexity?

$$low[v] = min(dfs[v], min_{(v,w)\in E} \begin{cases} low[w], & (v,w) tree-edge \\ dfs[w], & (v,w) rest-edge \end{cases})$$

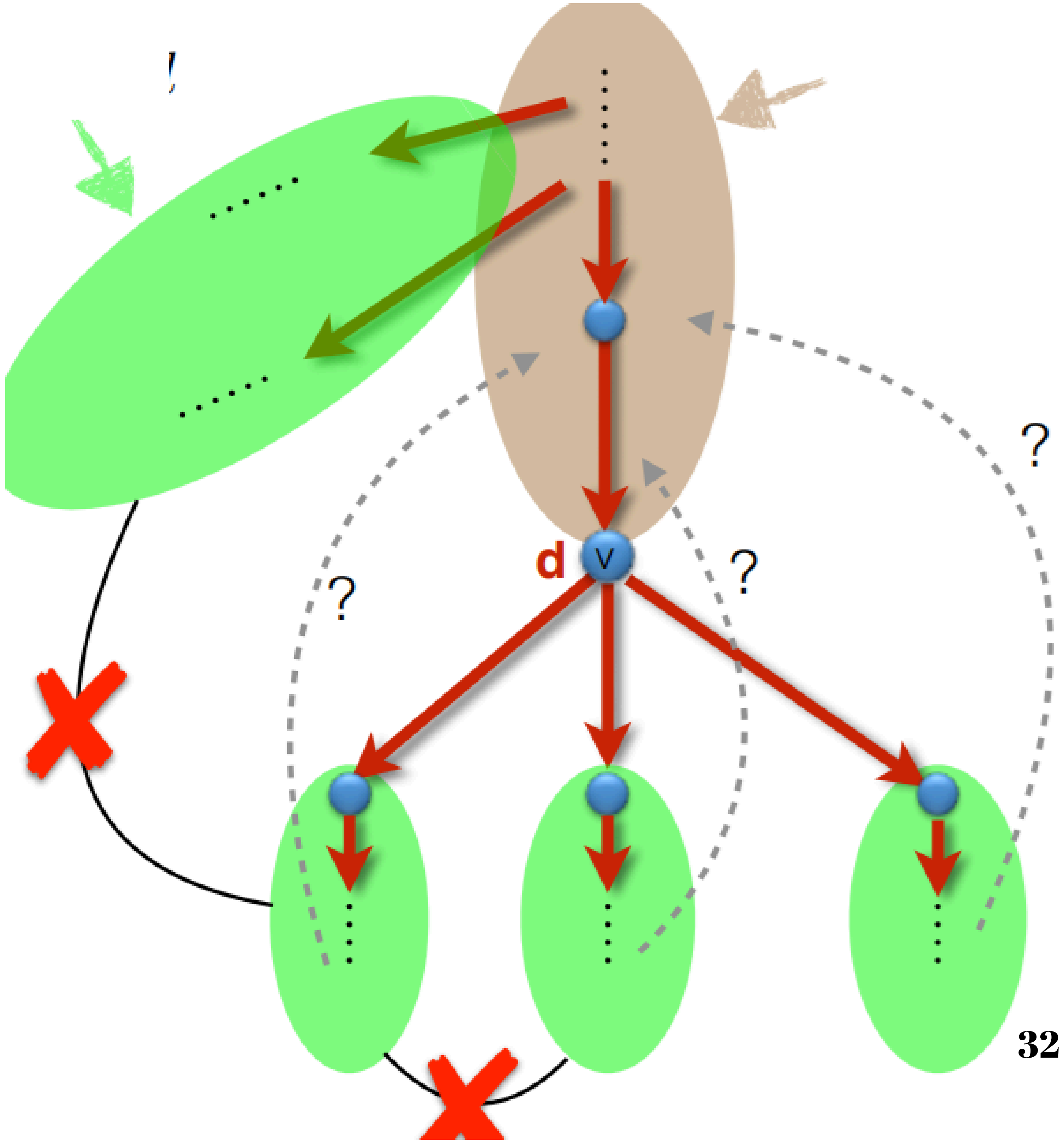
O(n + m) to compute low... same as for dfs-tree =>
total of O(n +m)

How can one use the low and dfs values in order to argue if a node is a cut vertex?

Case distinction :

a). v is a root and deg(v) in T >= 2 **why?**

*b). v is **not** the root and for some u - child of v, low[u] >= dfs[v]* **why?**

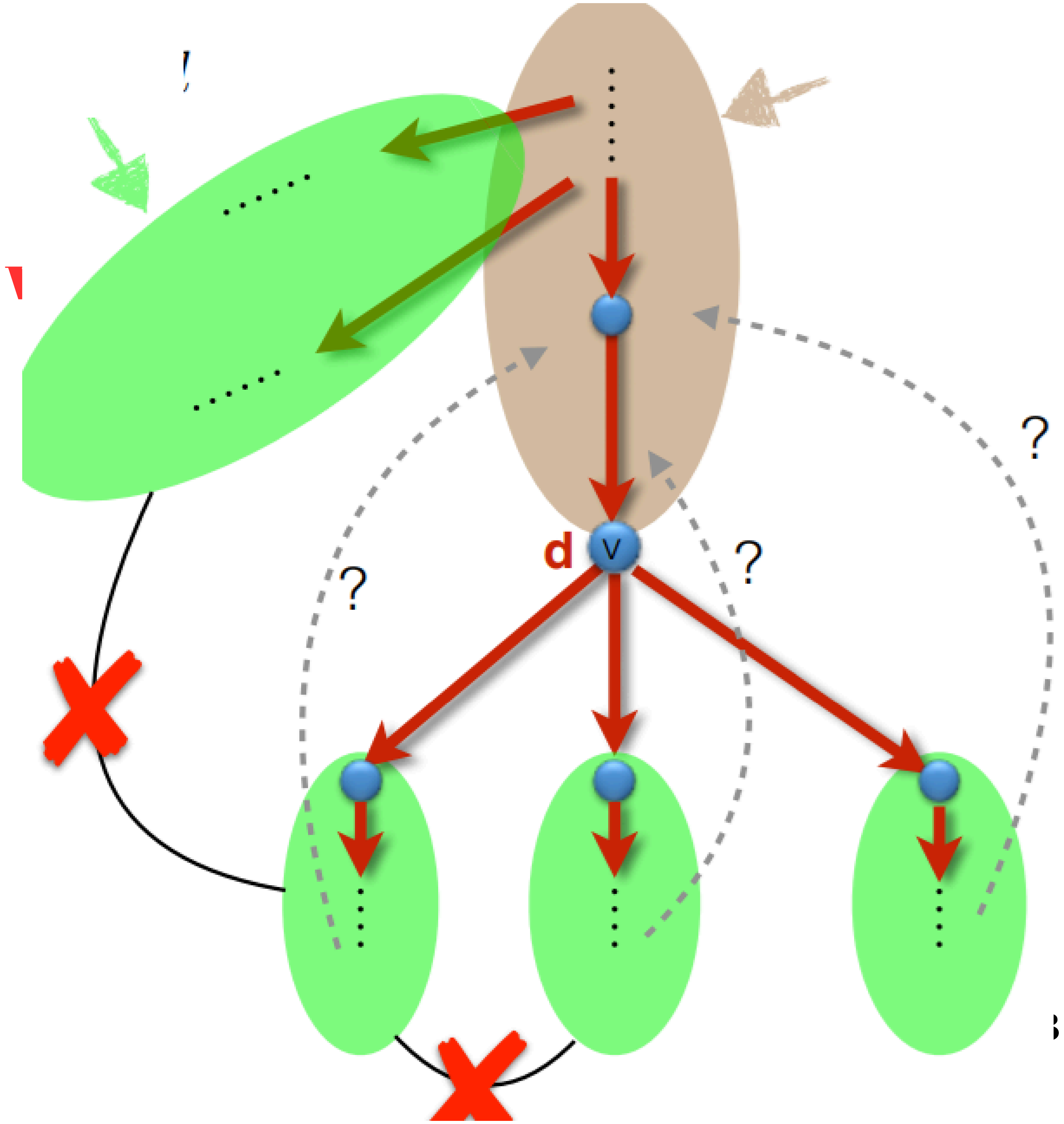


How can one use the low and dfs values in order to argue if a node is a cut edge?

Case distinction :

a). an unused edge (rest-edge) ***can't*** *be a cut edge* **why?**

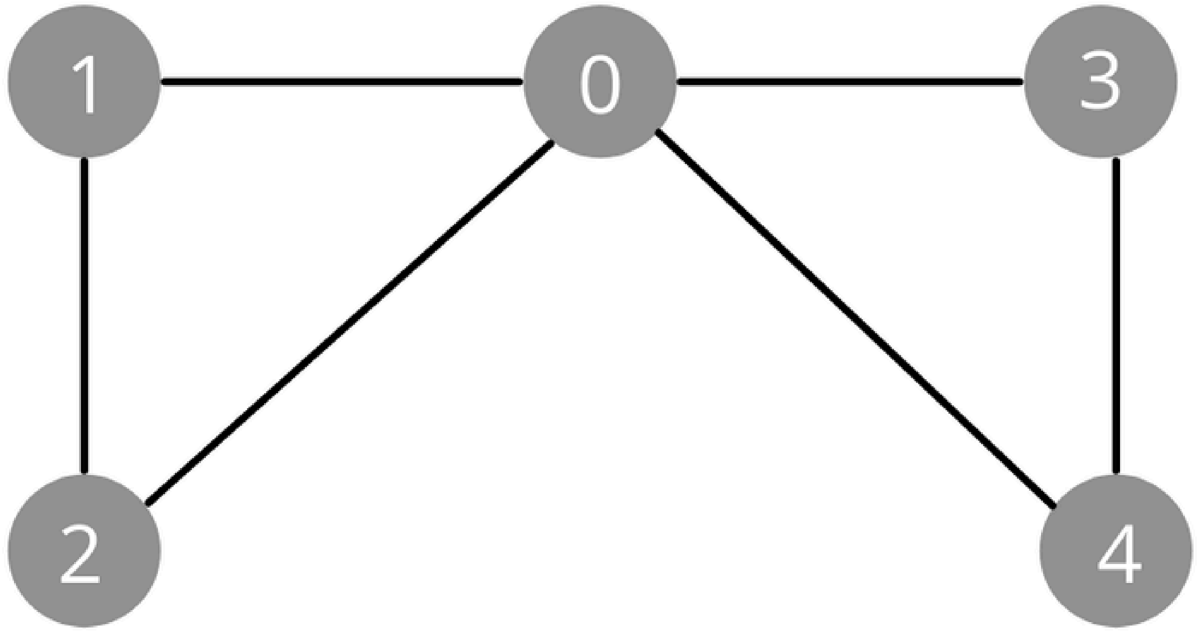
b). a tree-edge (v, w) is a cut edge iff low[w] > dfs[v] !! ***(why? why not >=? even possible?)***

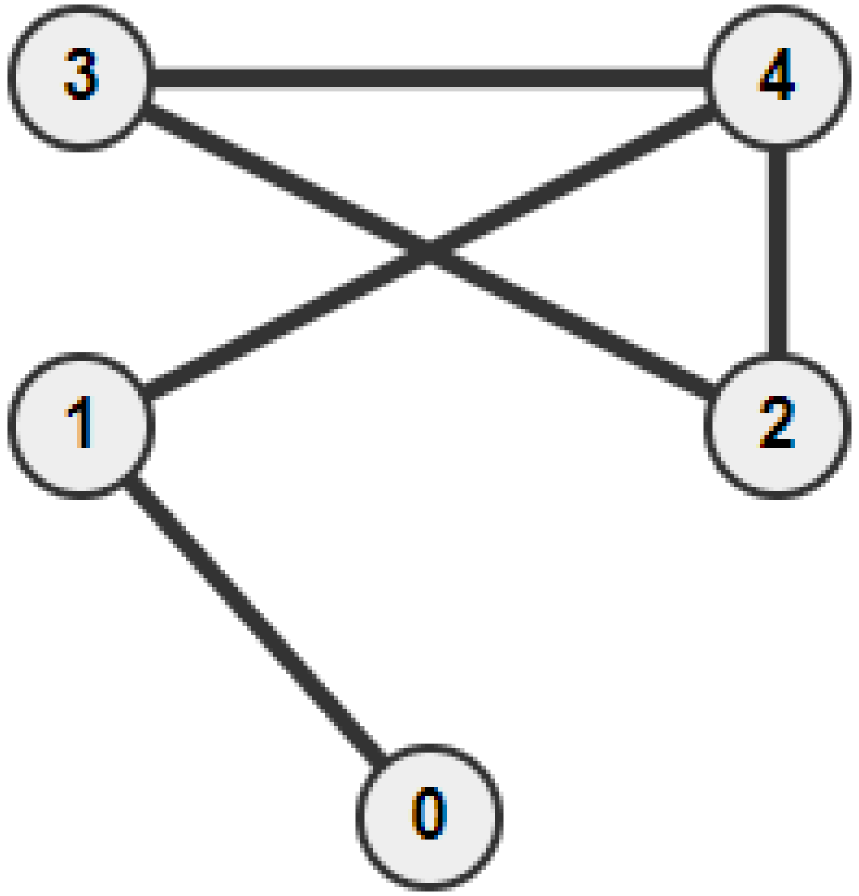


Special graphs

Eulerian graphs. Eulerian circuits. Eulerian paths

What is an Eulerian graph? => A graph that contains an Eulerian circuit => ….. def?





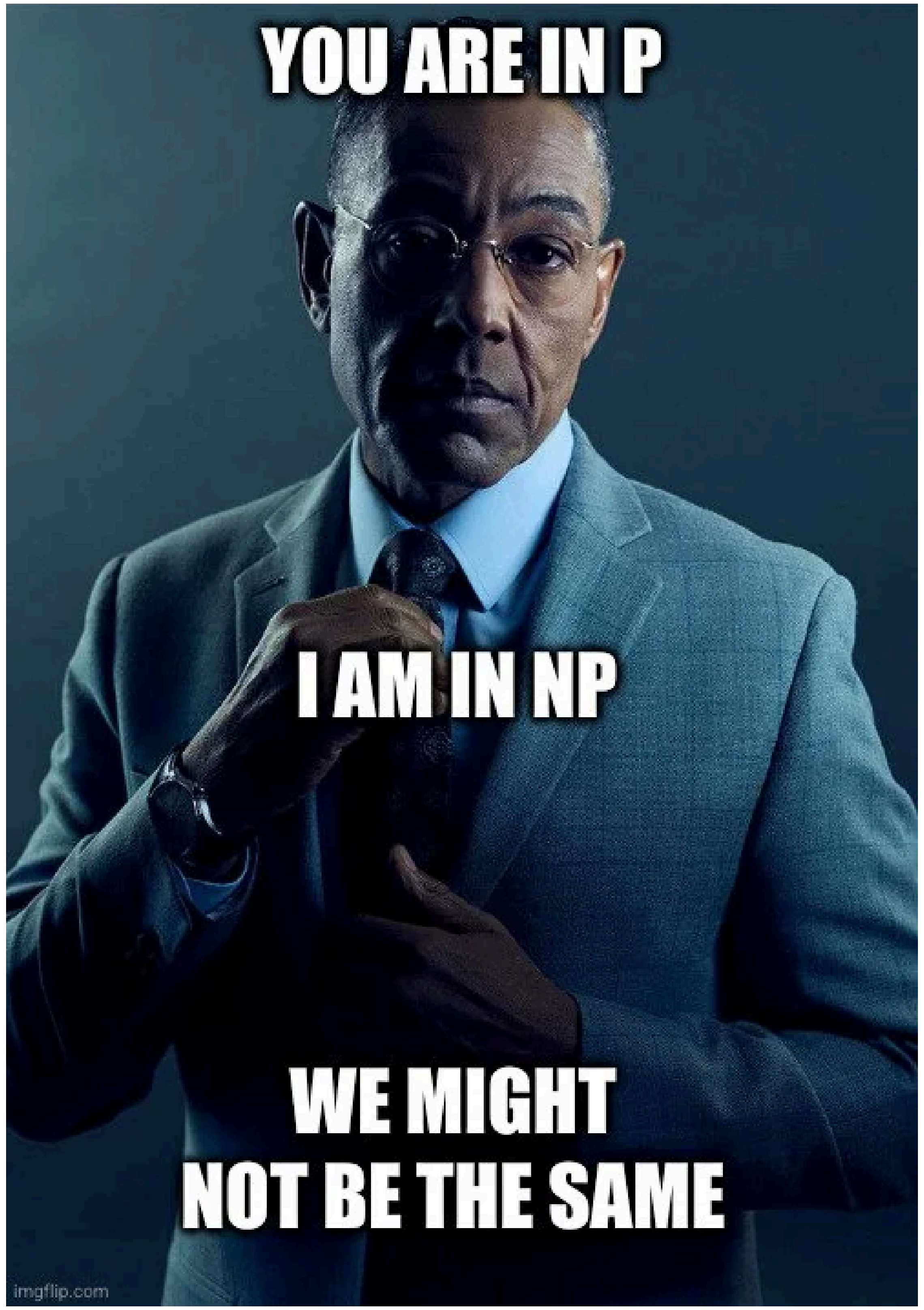
How do I know if a graph has an Eulerian Circuit?

How do I know if a graph has an Eulerian path?

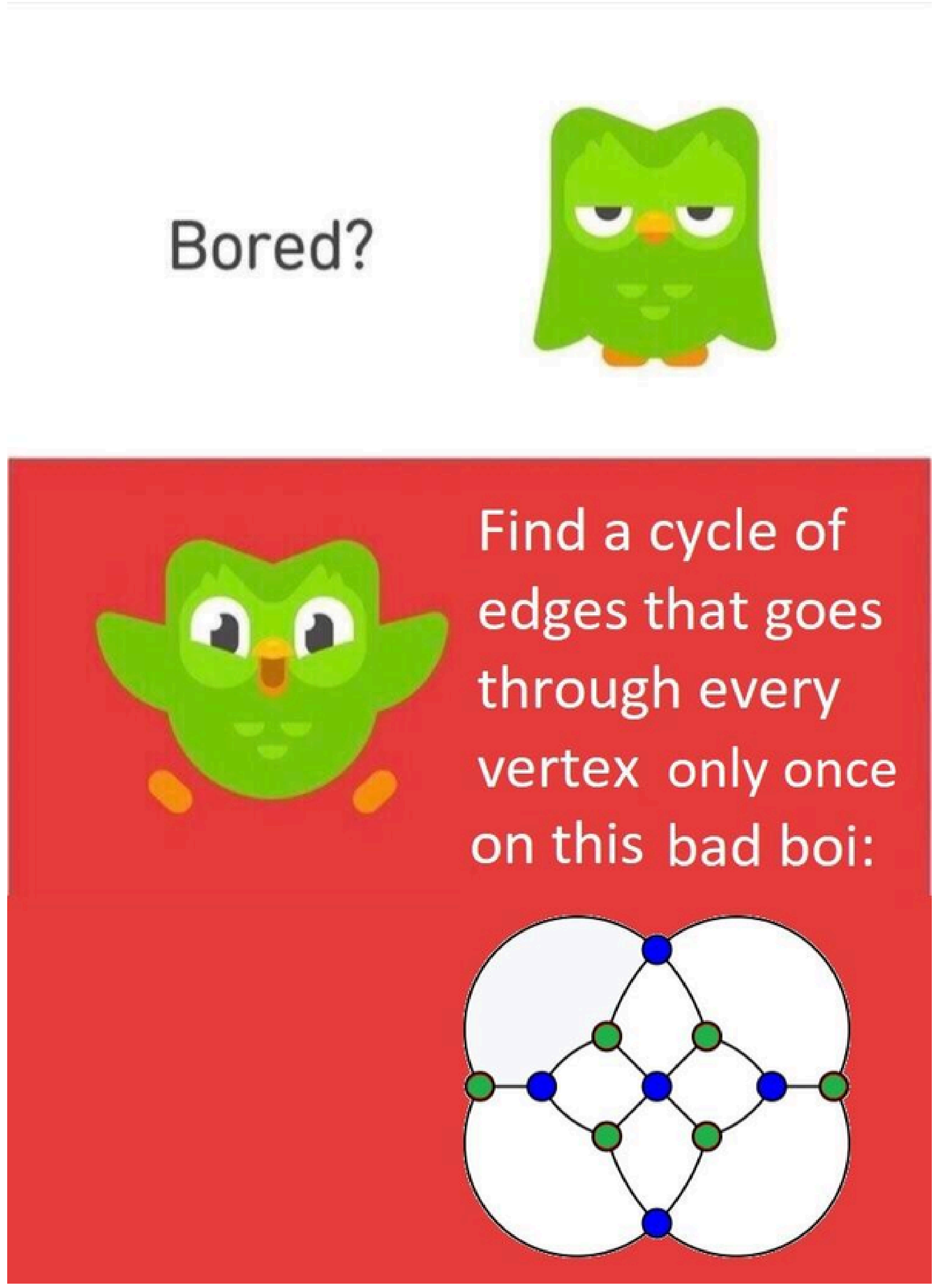
Eulerian graphs. Eulerian circuits. Eulerian paths

How can we actually find an Eulerian circuit?
... DFS-post numbers algorithms (from AnD course)...

It is just a slightly modified DFS (i.e. in order to save the post number) => O(n + m)... so it has a polynomial complexity

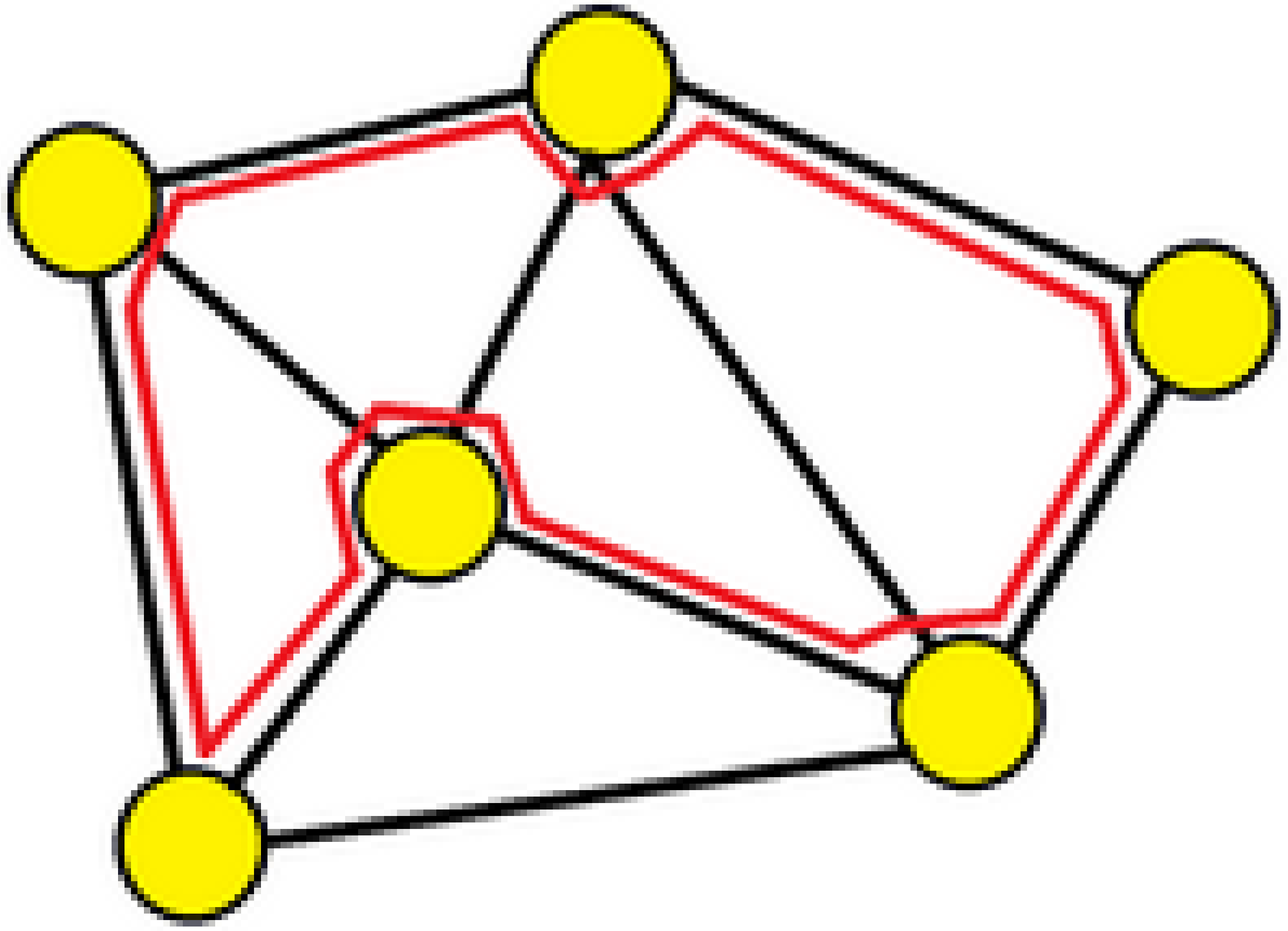


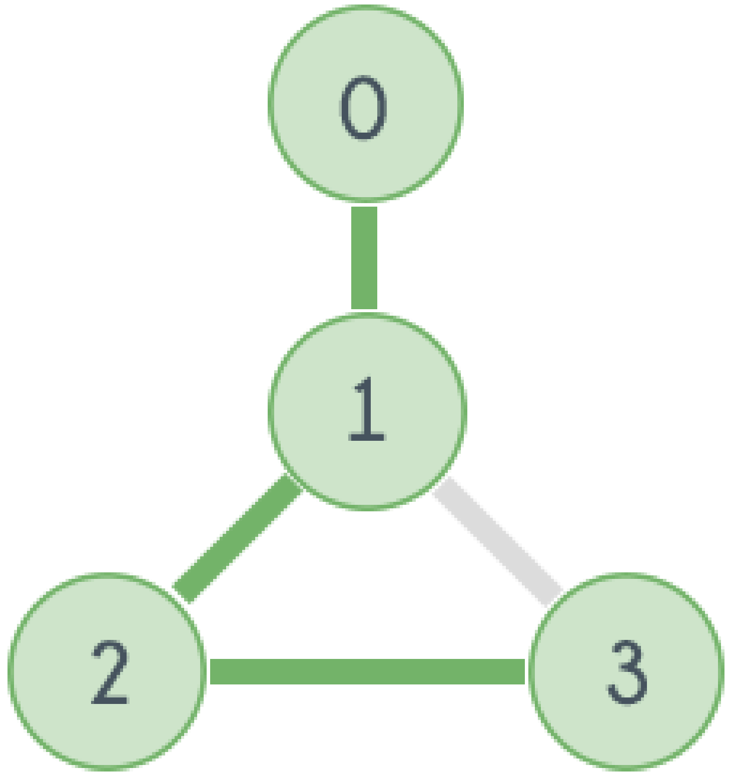
Hamiltonian Cycle. Hamiltonian path.

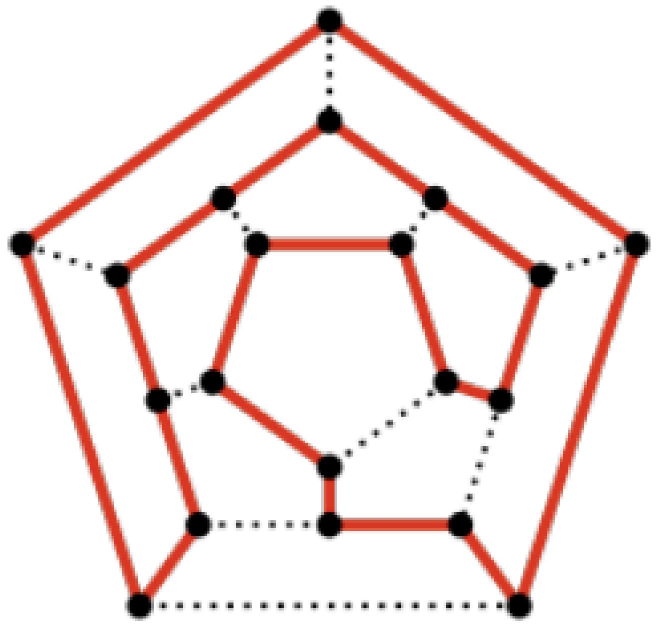


Hamiltonian Cycle. Hamiltonian path.

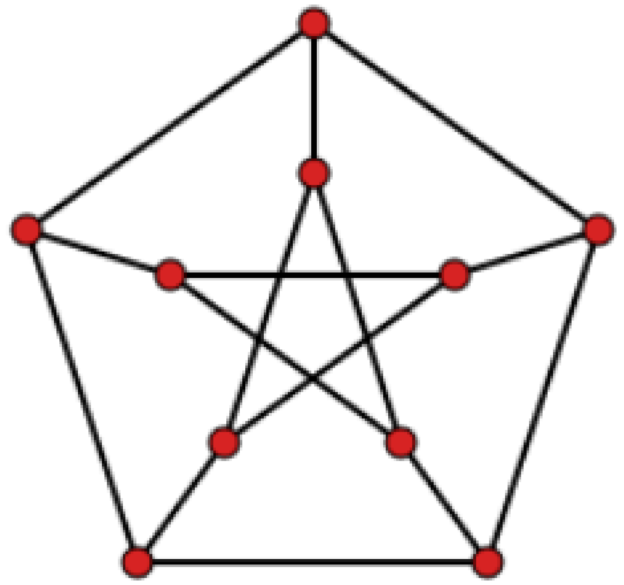
Definition? What are we actually searching?







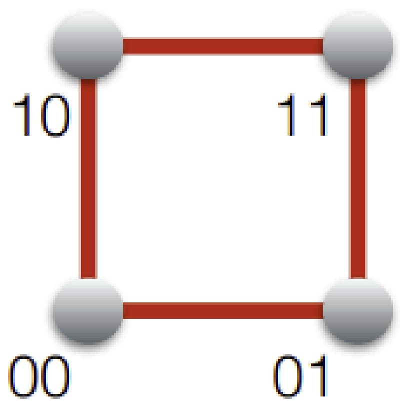
Ikosaeder



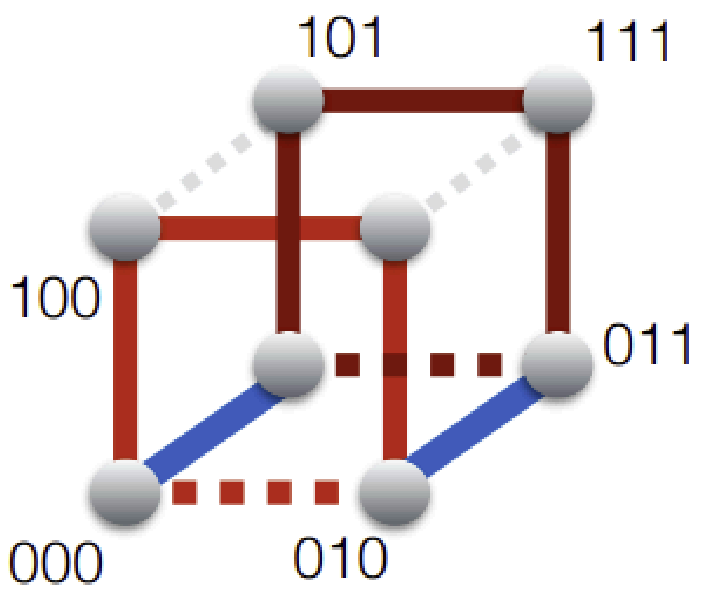
Petersengraph

Hamiltonkreis im Hyperwürfel: Gray-Code

d=2:



d=3:



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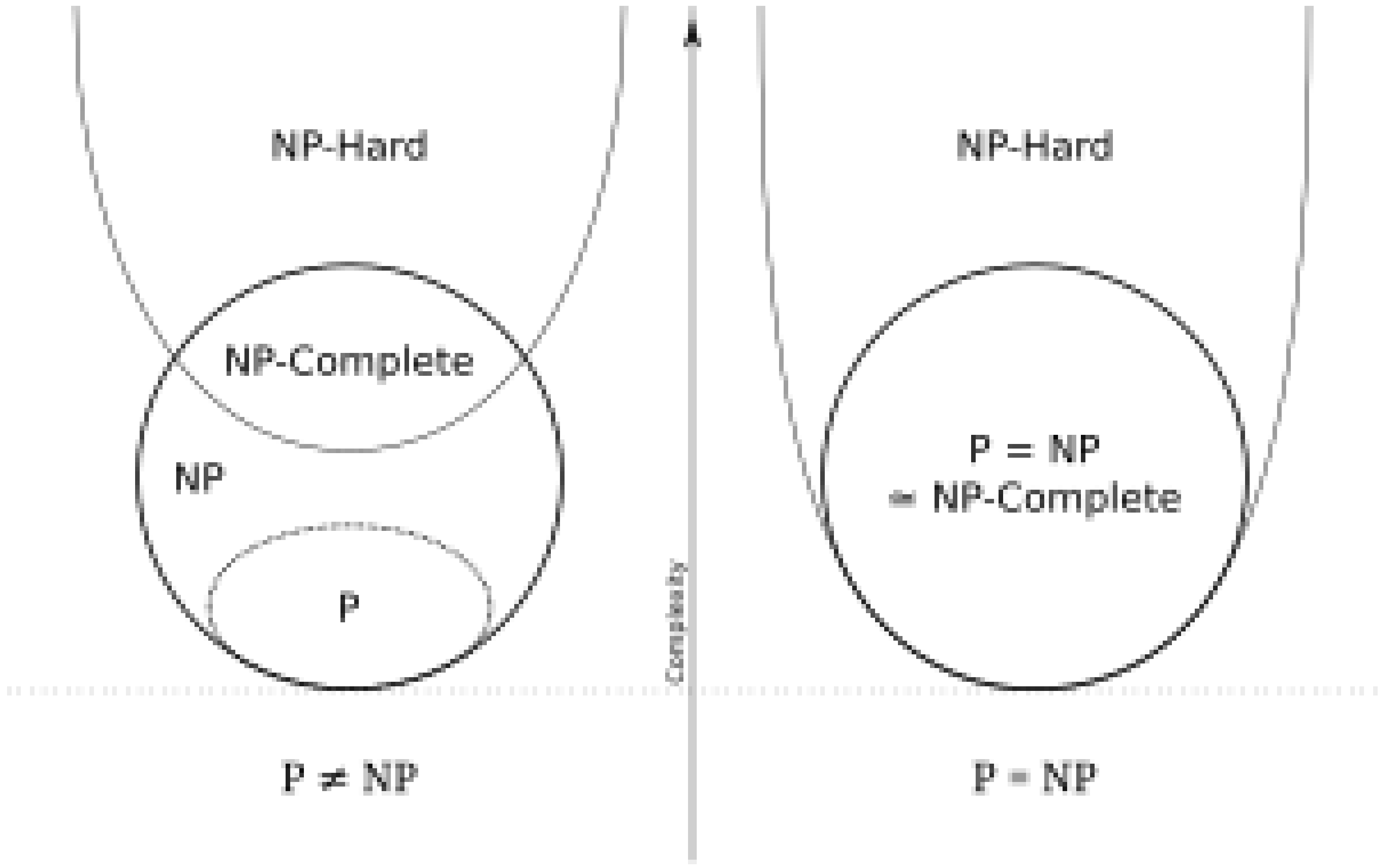
analog für d ≧ 4

Bad news… Finding an Hamiltonian Cycle / Path or even arguing if one exists is in NP …

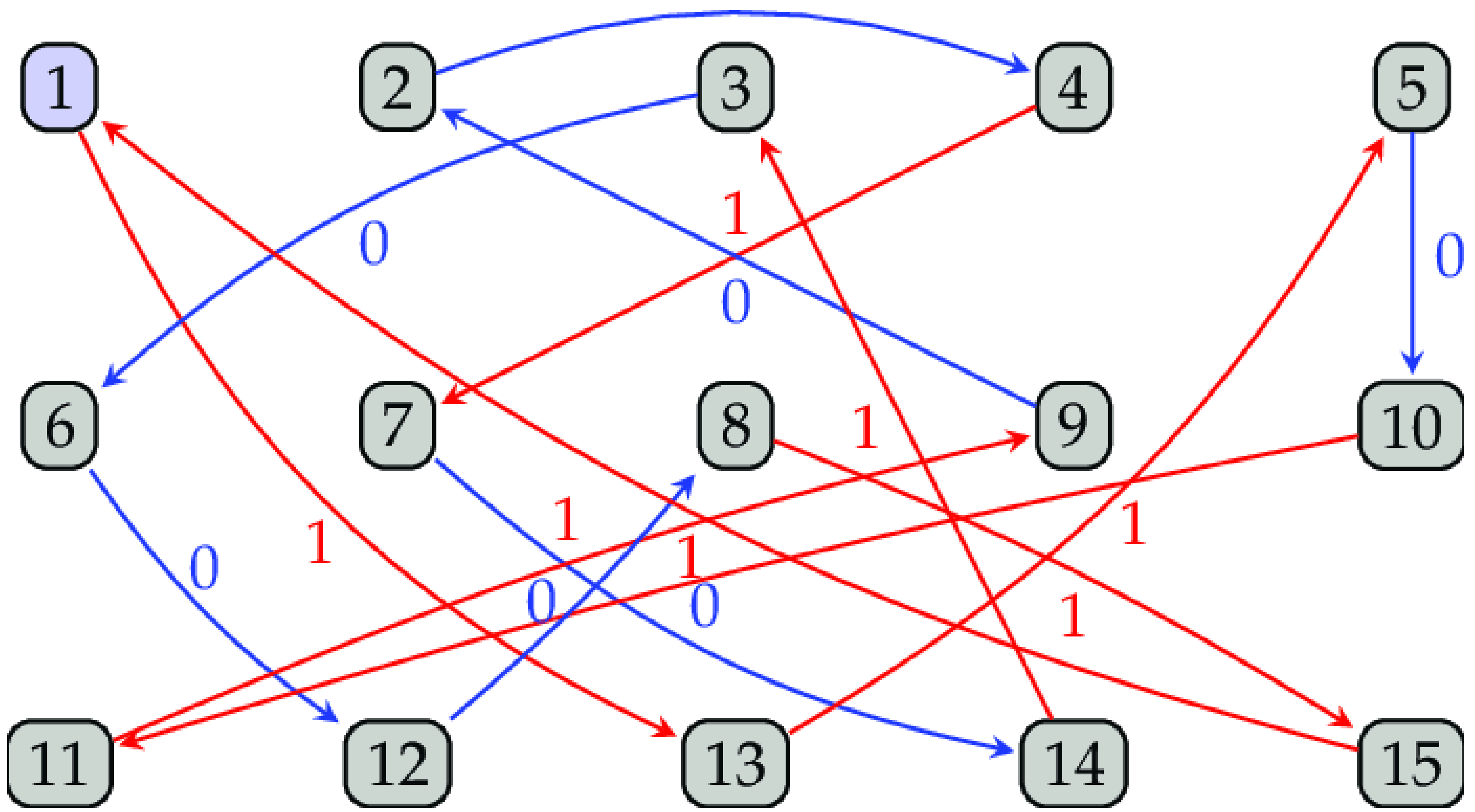
… i.e. there is no alg. in polynomial time that solves this problem… or at least under the idea of P != NP



Hamiltonian cycle

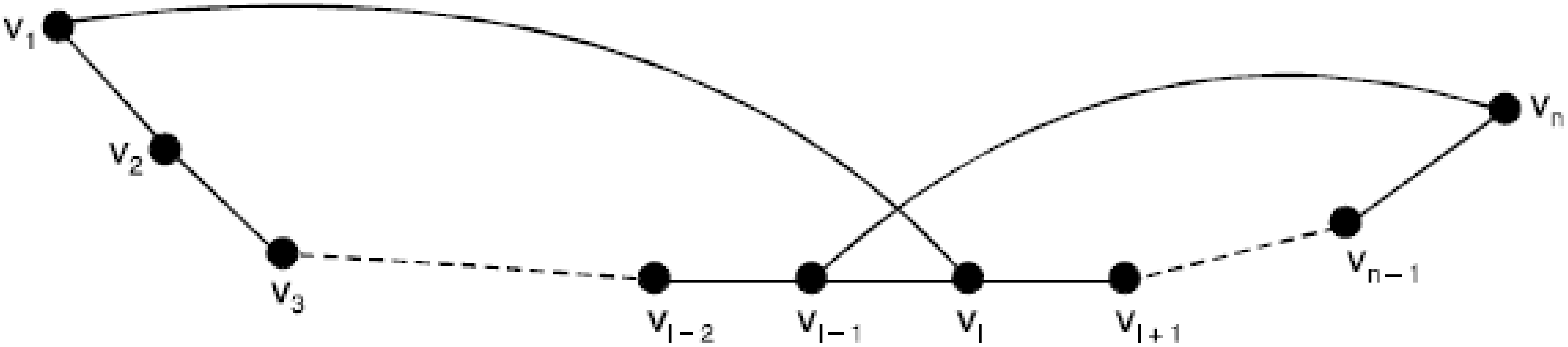
- *naive approach … O(n!) - Brute force - how?*
- *a better complexity - almost O(2^n) - how?*

DP approach… (whiteboard)



Dirac’s Theorem

Dirac's Theorem (1952) — A simple graph with n vertices ($n \geq 3$) is Hamiltonian if every vertex has degree $\frac{n}{2}$ or greater.



proof: (whiteboard)

Extra-exercises

Der Kreis auf acht Knoten (C_8) ist 2-zusammenhängend.

Bitte wählen Sie eine Antwort:

Der Kreis auf acht Knoten (C_8) ist 2-zusammenhängend.

Bitte wählen Sie eine Antwort:

- Wahr
- Falsch

In einer Tiefensuche sei $w \in V$ das erste Kind von $v \in V$ im DFS-Baum.

Falls $\text{low}[w] \geq \text{dfs}[v]$, so ist v ein Artikulationsknoten.

Bitte wählen Sie eine Antwort:

In einer Tiefensuche sei $w \in V$ das erste Kind von $v \in V$ im DFS-Baum.

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Bitte wählen Sie eine Antwort:

- ○ Wahr
- ◉ Falsch ✓

Vollständige Graphen sind die einzigen zusammenhängenden Graphen, welche sowohl eine Eulertour als auch einen Hamiltonkreis enthalten.

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Exercise 1 – *Connectivity*

(a) Assume G is 2-connected. Let (u, v, w) be a path in G of length 2. Show that we can extend this path to a cycle, that is, show that G contains a cycle where u, v, w appear as incident vertices.

(b) Prove that if $G = (V, E)$ is a 2-connected graph, $e \in E$, and $u, v \in V$ is a pair of vertices in G not incident to e, then there is a path in G from u to v passing through e. (*Hint:* in a 2-connected graph, every two edges e, f lie on a common cycle.)